NVIDIA Driver Settings Guide

Every setting in NVIDIA Control Panel, in the right order, with a screenshot for each step. The screenshots are an example from driver 582.78 on a GeForce GTX 1070; the advice applies to other GeForce cards too.

Part of Windows 11 Optimizer

In a hurry?
Open "Quick setup". Nine short steps, about 10 minutes.

Want to see each screen?
Open "Step by step". Every page has a screenshot with numbered markers.

Something looks wrong?
Open "Troubleshooting". Find the symptom and the fix.

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Before you start

Five minutes of reading that save you time later.

What this guide covers, and what it does not

It covers **NVIDIA Control Panel**, the driver settings that NVIDIA itself provides, plus the Windows and monitor settings that go with them. It never edits a game file or hooks into a game, so anti-cheat systems do not treat it as cheating.

The example used in this guide

Part	The screenshots show
Graphics card	GeForce GTX 1070 (Pascal). The same pages exist on GTX 10, GTX 16 and RTX cards, so you can follow this guide with another card. If an option is missing on yours, skip it. Newer cards may have extra options: see "Other cards and newer drivers".
Driver	Driver 582.78. Newer drivers look almost the same; rows can move or be added.
Windows	Windows 11.
Screen	A 144 Hz Adaptive-Sync (FreeSync) monitor connected with a **DisplayPort** cable, used as G-SYNC Compatible.

About the example driver, 582.78

News sites report that 582.78 is a **GeForce Security Update Driver** for GeForce GTX 750 to GTX 10 series cards (Maxwell, Pascal and Volta), published at the end of September 2026. NVIDIA stopped Game Ready drivers for these cards in October 2025 and has announced security updates until October 2028. In plain words:

- The driver brings **no new game optimizations**. Nothing in this guide depends on a newer driver.
- If your games run well, **you do not need to update the driver every month**. Update when a security update appears.
- The Control Panel pages look the same as before. To see your version, click **System Information** at the bottom left of NVIDIA Control Panel: the **Display** tab shows "Driver version".

These facts come from news coverage. NVIDIA's own driver page and release notes are the authority: check nvidia.com if the exact wording matters to you.

Honest expectations

Most options change FPS by 0 to 3%. The ones that really change how a game feels are **power management mode, low latency mode, V-Sync / G-SYNC / frame cap** and the **refresh rate**. In-game settings and your CPU matter more than all driver options together.

Three safety habits

- Take a screenshot of each page before you change it (Windows key + Shift + S).
- **Restore Defaults** at the top right of a page puts that page back to the NVIDIA defaults.
- Nothing is saved until you press **Apply** at the bottom right of the page.

Quick setup in 9 steps

The short path. Each step points to the page with the screenshot.

1

Open the Control Panel, then image settings

Right-click an empty spot on the desktop, **Show more options**, **NVIDIA Control Panel** (if it is missing, install it from the Microsoft Store). Then 3D Settings, **Adjust image settings with preview**. Pick **Use the advanced 3D image settings**, then Apply.

Why: the slider preset silently overrides your own values.

2

Manage 3D settings: the nine values that matter

Open **Global Settings** and set these. Leave the other rows at their defaults (pages "2a" to "2c" list every row).

Row	Set to
Low Latency Mode	On
Power management mode	Prefer maximum performance
Preferred refresh rate	Highest available
Shader Cache Size	Unlimited (or 10 GB)
Texture filtering - Negative LOD bias	Clamp
Threaded optimization	Auto
Monitor Technology	G-SYNC Compatible (if the row exists)
Vertical sync	Off for style A, On for style B
Max Frame Rate	Off for style A, refresh rate minus 3 for style B

3

PhysX: name your card

3D Settings, **Configure Surround, PhysX**. Processor: your GeForce card, not Auto-select. Apply.

4

Resolution and refresh rate

Display, **Change resolution**. Native resolution, the highest refresh rate. Under "Use NVIDIA color settings": RGB, 8 bpc, Full range. Apply.

5

Desktop size and position

Display, **Adjust desktop size and position**. If you play at native resolution you can leave it. Otherwise pick **Aspect ratio** and leave "Override the scaling mode" unticked.

6 G-SYNC

Display, **Set up G-SYNC**. Tick **Enable G-SYNC, G-SYNC Compatible**, choose **Enable for full screen mode**, tick **Enable settings for the selected display model**. Apply.

7 Make your gaming screen the main display

If the G-SYNC page says "set it as primary display": Windows Settings, System, Display, select the gaming screen, tick **Make this my main display**.

8 Inside the game

Fullscreen (or borderless), V-Sync **Off** in the game for style B, NVIDIA Reflex **On** if the game offers it.

9 Test it

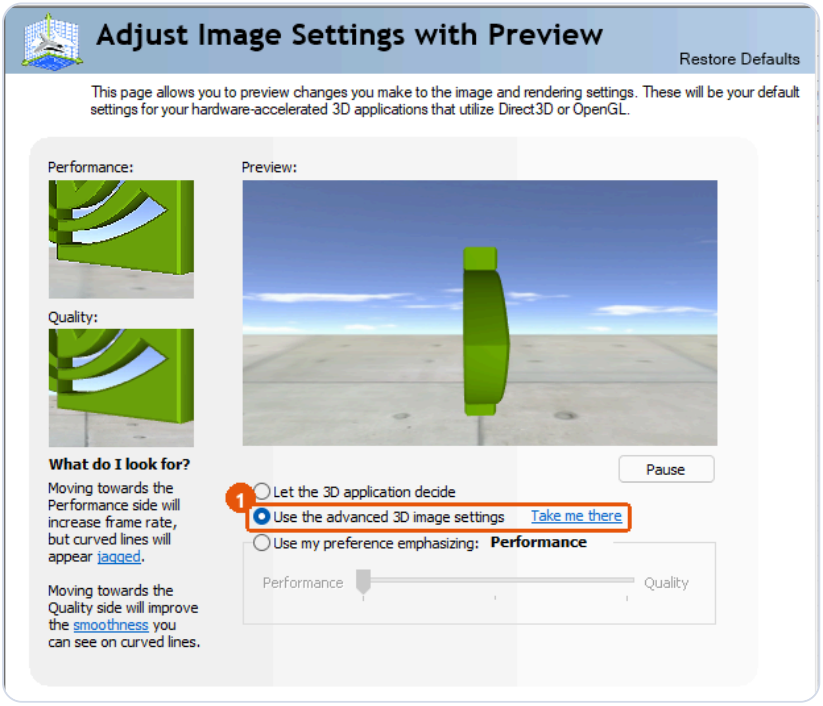
Play five minutes in a game you know. Is the picture torn? Does it stutter? If yes, use the Troubleshooting page, or press **Restore Defaults** on the page you changed last.

Step by step, with screenshots

The orange numbers on each screenshot match the numbered points under it.

Step 1. Adjust image settings with preview

Path: 3D Settings, Adjust image settings with preview.



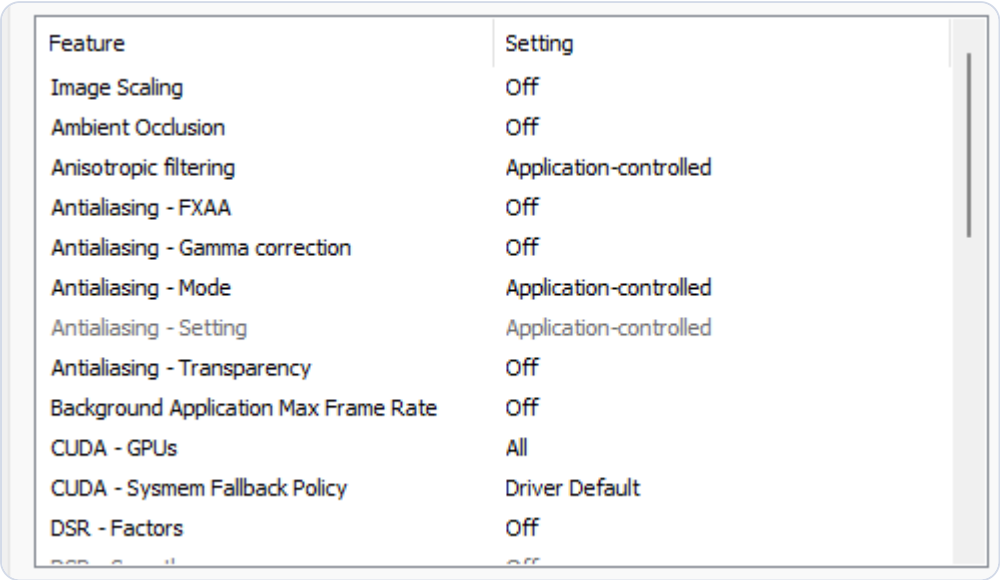
Driver 582.78, GTX 1070. Your page looks the same.

1. Select **Use the advanced 3D image settings**. The slider underneath turns grey, which is what you want.
2. Press **Apply** at the bottom right.

Why not the slider? "Use my preference emphasizing" is a preset that changes image-quality options in the background and overrides the values you set on the next page. The speed difference is about 1 to 2%.

Step 2a. Manage 3D settings: the top of the list

Path: 3D Settings, Manage 3D settings, tab **Global Settings**. Click a value in the right column to open its drop-down list, then press **Apply** when the whole page is done.

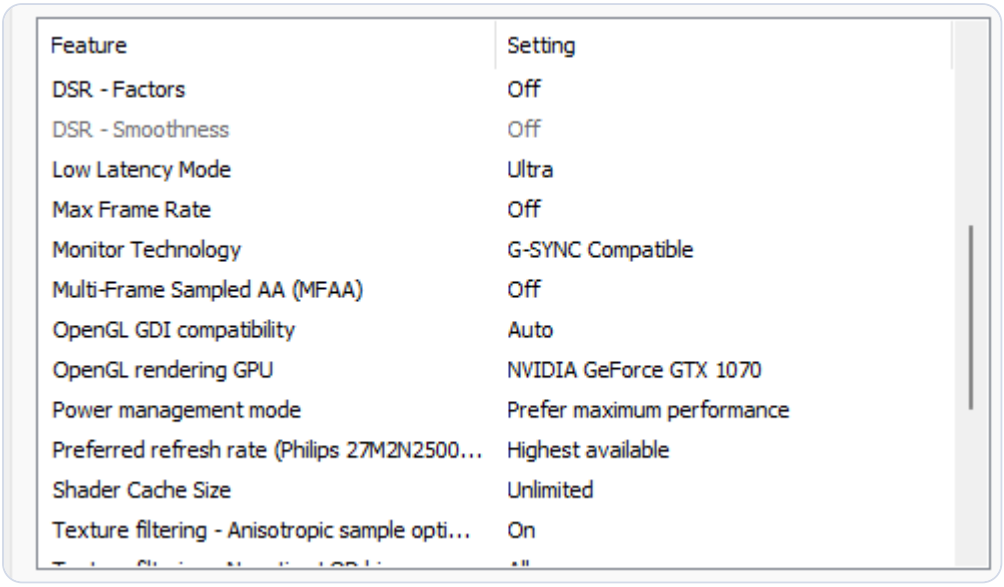


Global Settings, top of the list.

Row	Set to	In one line
Image Scaling	Off	NVIDIA's upscaler. Only for playing below native resolution.
Ambient Occlusion	Off	Forces extra shadows. Big FPS cost, few games support it.
Anisotropic filtering	Application-controlled	Let the game choose. Forcing 16x costs 0 to 3% and helps old games.
Antialiasing - FXAA	Off	Blur-based edge smoothing. Games have better options.
Antialiasing - Gamma correction	Leave as is	Matters only when the driver forces antialiasing.
Antialiasing - Mode	Application-controlled	Let the game decide. "Override" rarely does anything in modern games.
Antialiasing - Setting	Nothing to do	Greyed out while Mode is Application-controlled.
Antialiasing - Transparency	Off	Smooths fences and leaves. Supersampling is expensive.
Background Application Max Frame Rate	Off	Caps a game that is in the background. 20 saves heat and power.
CUDA - GPUs	All	Which cards CUDA programs may use. Not for games.
CUDA - Sysmem Fallback Policy	Driver Default	What happens when video memory is full.
DSR - Factors	Off	Renders above native resolution and shrinks it. Lowers FPS on purpose.

Step 2b. Manage 3D settings: the middle of the list

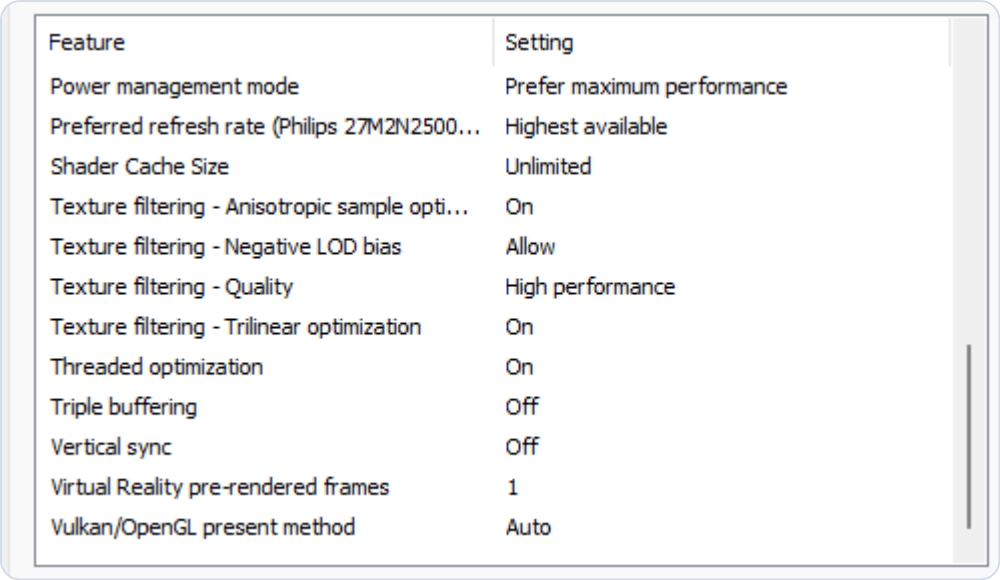
Scroll the list down with the mouse wheel or the scroll bar at its right edge.



Global Settings, middle of the list.

Row	Set to	In one line
DSR - Smoothness	Nothing to do	Greyed out while DSR is off.
Low Latency Mode	**On**	Shortens the render queue. Ultra for GPU-bound games: see "Choose your style".
Max Frame Rate	**Style A: Off.** Style B: refresh rate minus 3	The driver's frame cap. For 144 Hz, 141.
Monitor Technology	**G-SYNC Compatible**	The screen follows your FPS. Shown only after step 6.
Multi-Frame Sampled AA (MFAA)	Off	Rarely useful and needs MSAA.
OpenGL GDI compatibility	Auto	For very old OpenGL programs.
OpenGL rendering GPU	Your GeForce card	Name the card so the driver does not guess.
Power management mode	**Prefer maximum performance**	Keeps clocks steady while a game runs. Costs a little heat and idle power.
Preferred refresh rate	**Highest available**	Stops games from picking 60 Hz on a 144 Hz screen.
Shader Cache Size	**Unlimited** (or 10 GB)	Room for compiled shaders, so fewer stutters the second time.
Texture filtering - Anisotropic sample optimization	Off (On for esports)	Fewer texture samples: a tiny speed gain, a tiny quality loss.

Step 2c. Manage 3D settings: the bottom of the list



Feature	Setting
Power management mode	Prefer maximum performance
Preferred refresh rate (Philips 27M2N2500…	Highest available
Shader Cache Size	Unlimited
Texture filtering - Anisotropic sample opti…	On
Texture filtering - Negative LOD bias	Allow
Texture filtering - Quality	High performance
Texture filtering - Trilinear optimization	On
Threaded optimization	On
Triple buffering	Off
Vertical sync	Off
Virtual Reality pre-rendered frames	1
Vulkan/OpenGL present method	Auto

Global Settings, bottom of the list.

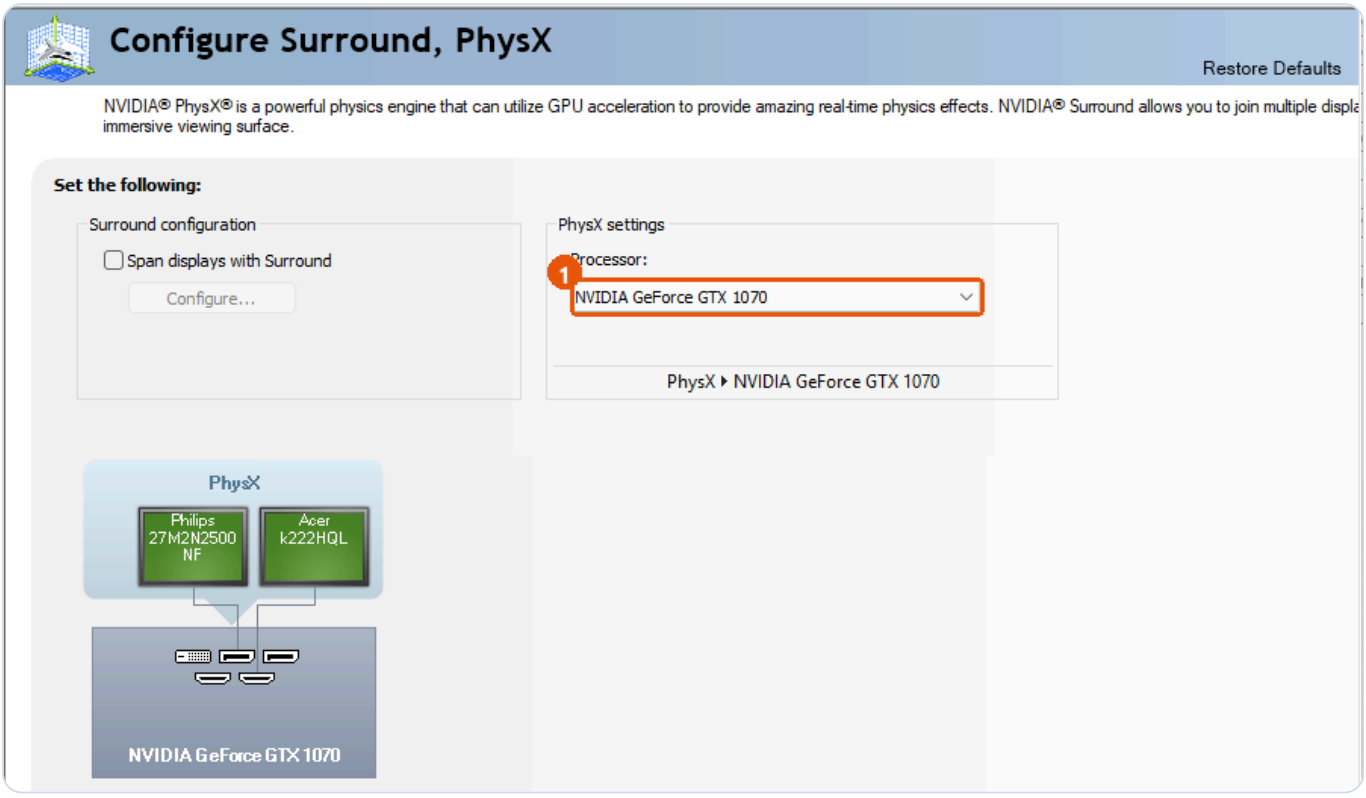
Row	Set to	In one line
Texture filtering - Negative LOD bias	**Clamp**	Removes texture shimmer. No FPS effect.
Texture filtering - Quality	Quality (High performance for esports)	Quality for looks, High performance for the last frames.
Texture filtering - Trilinear optimization	On	Faster texture blending, tiny loss. Leave it.
Threaded optimization	**Auto**	The driver chooses per game. Force On or Off for one game only if it misbehaves.
Triple buffering	Off	Only for OpenGL games with V-Sync.
Vertical sync	**Style A: Off.** Style B: On	See "Choose your style".
Virtual Reality pre-rendered frames	1	For VR headsets only.
Vulkan/OpenGL present method	Auto	Leave Auto unless a guide for one game says otherwise.

Press Apply, then check

Press **Apply** at the bottom right. Your screenshot may show other values than the "Set to" column. Follow the column. "Every setting explained" near the end of this guide has the long version of each row.

Step 3. Configure Surround, PhysX

Path: 3D Settings, Configure Surround, PhysX.



PhysX settings, processor drop-down (marker 1).

1. Open the **Processor** drop-down (marker 1) and choose your GeForce card instead of Auto-select. Auto normally picks the card anyway; naming it only removes the chance of physics work landing on the CPU.
2. Leave "Span displays with Surround" unticked. Surround is for three-screen setups.
3. Press **Apply**.

Step 4. Change resolution

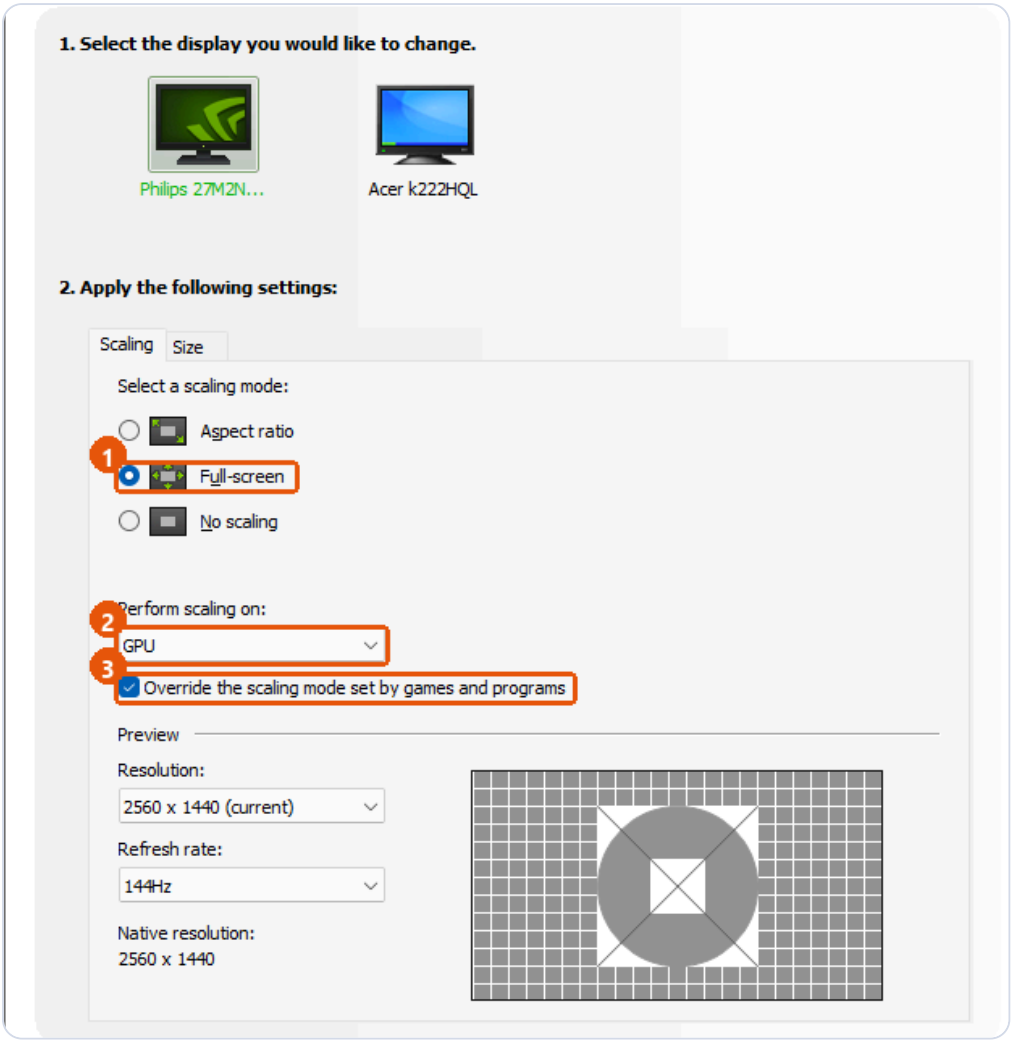
Path: Display, Change resolution. This page has no screenshot here because it lists your own screens.

1. Select the gaming screen at the top.
2. **Resolution:** the one marked "native" (2560 x 1440 for the screen in the screenshots).
3. **Refresh rate:** the highest value (144 Hz). A 144 Hz screen left at 60 Hz is the most common reason for "my FPS feels low".
4. Under **Choose the color settings** select **Use NVIDIA color settings** and set **Output color format: RGB**, **Output color depth: 8 bpc**, **Output dynamic range: Full**. Most games and Windows use 8 bits, and a higher depth gives no benefit with them. Full range keeps blacks black (Limited is for TVs).
5. Press **Apply**. Windows shows "Keep changes?": press **Yes**.

> ⚠ **If the screen goes black**
>
> Do nothing for 15 seconds. Windows returns to the old mode by itself if you do not confirm.

Step 5. Adjust desktop size and position

Path: Display, Adjust desktop size and position.



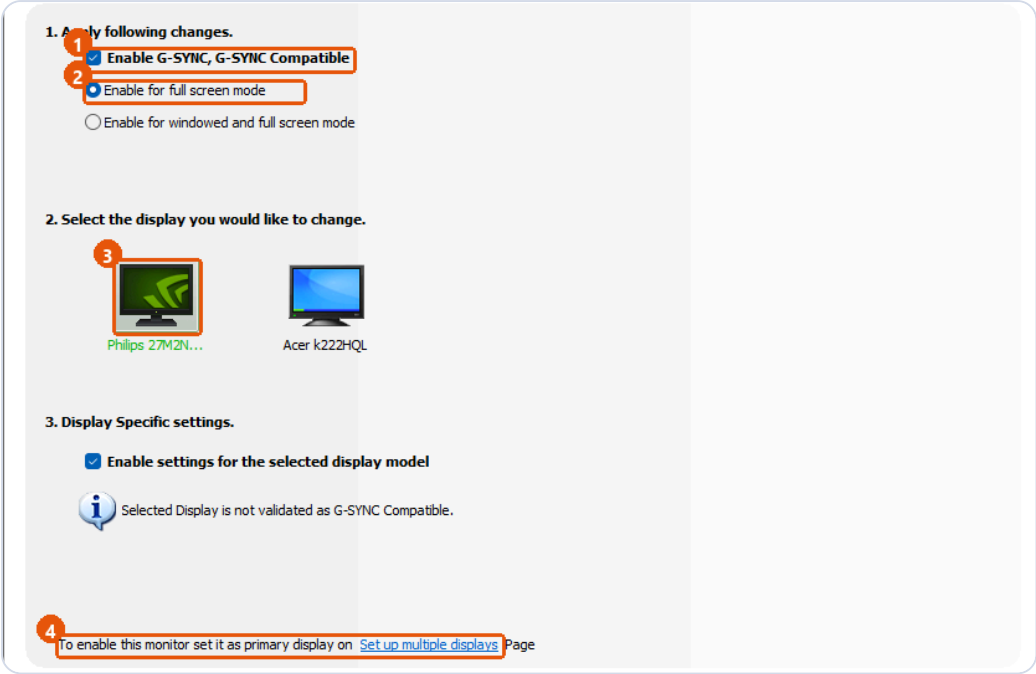
Tab Scaling. Markers 1 to 3 are the three choices explained below.

1. **Scaling mode** (marker 1). When the game runs at your screen's native resolution, the mode makes no difference. When it runs lower, **Aspect ratio** keeps the picture the right shape. Full-screen stretches it. No scaling leaves black borders.
2. **Perform scaling on** (marker 2). **GPU** is the safe default. **Display** lets the monitor do the scaling, which is a little faster on some screens and worse looking on others. Try it only if you play below native resolution.
3. **Override the scaling mode set by games and programs** (marker 3). Leave it **unticked**, so games can use their own scaling. Tick it only if one game shows a stretched picture.

> **Rule of thumb**
>
> Play at native resolution and this page needs no change at all.

Step 6. Set up G-SYNC

Path: Display, Set up G-SYNC.



Set up G-SYNC with a monitor that is not NVIDIA-validated.

1. Tick **Enable G-SYNC, G-SYNC Compatible** (marker 1). This lets the screen follow your FPS, so you get no tearing without the delay of V-Sync.
2. Choose **Enable for full screen mode** (marker 2). It covers fullscreen and the fullscreen-style windows of most games. "Windowed and full screen" also works for windowed games.
3. Select your gaming screen (marker 3), then tick **Enable settings for the selected display model** (section 3 of the page).
4. Read the line under it. **"Selected Display is not validated as G-SYNC Compatible"** means NVIDIA has not tested that model. It often works, but NVIDIA does not promise a perfect result. If you see flicker or black flashes, untick the box again.
5. Marker 4: **"To enable this monitor set it as primary display"**. If you see this line, open the **Set up multiple displays** link, or Windows Settings, System, Display, select the gaming screen and tick **Make this my main display**. Press Apply on both pages.

> ⚠ **Use DisplayPort**
>
> With a GTX 10 series card, G-SYNC Compatible needs a DisplayPort connection (DisplayPort 1.2a or newer). HDMI variable refresh rate needs a newer card than Pascal, according to hardware sites. If your screen is on HDMI, move the cable to DisplayPort.

Step 7. Program Settings: different values for one game

Global Settings apply to everything. **Program Settings** apply to one game and win over Global. Use it for the one or two games that need something else (for example Low Latency Mode Ultra for a GPU-heavy game).

1. 3D Settings, Manage 3D settings, tab **Program Settings**.
2. Press **Add** and pick the game's .exe (or choose it from the list).
3. Change only the rows you need. Everything else stays **Use global setting**.
4. Press **Apply**. To undo, set the row back to "Use global setting", or press **Restore**.

Choose your style

V-Sync and the frame cap set the feel of a game more than any other option.

Two ready styles

Where	Style A: lowest delay	Style B: smooth, no tearing
For	Esports and shooters where every millisecond counts	Single-player and most other games
NVIDIA: Vertical sync	Off	**On**
NVIDIA: Max Frame Rate	Off (or a cap well above your refresh rate)	Your refresh rate minus 3 (144 Hz: **141**)
NVIDIA: Low Latency Mode	On	On
NVIDIA: Set up G-SYNC	On (the screen still follows your FPS below the refresh rate)	On
In the game: V-Sync	Off	**Off** (the Control Panel does the work)
In the game: NVIDIA Reflex	On, if offered	On, if offered
Price	Tearing (a horizontal split) on some frames	FPS cannot go above the cap

Why 3 FPS below the refresh rate?
Blur Busters, who tested G-SYNC in detail, recommend a frame cap a few FPS below the refresh rate (about 3 to 5%) together with V-Sync on in the Control Panel. The game then always stays inside the G-SYNC range, so V-Sync never adds its usual delay. It is community-tested guidance, not an NVIDIA rule. If the game has its own FPS limit, use that one first: it adds the least delay.

Low Latency Mode: On or Ultra?

Situation	Choose
The graphics card is the limit (GPU usage 95 to 99%): heavy, good-looking games	**Ultra** for that game (Program Settings). It cuts the render queue.
The CPU is the limit (GPU usage clearly below 95%): esports, very high FPS	**On** (Global). Ultra gives little and can cost a few FPS.
The game has NVIDIA Reflex	Turn **Reflex On** in the game and leave the driver at On. Reflex does the same job better.
DirectX 12 or Vulkan game	The driver setting does nothing there. Use Reflex if offered.
G-SYNC with V-Sync (style B) and Ultra	Ultra also keeps FPS under the refresh rate by itself. Fine, but then you do not need the cap.

How to see which limit you have: open Task Manager, Performance, GPU while you play. It needs no overlay and nothing hooks into the game. GPU at 95% or more means the card is the limit. Lower means the CPU or the game is.

Other cards and newer drivers

The screenshots are an example from driver 582.78. Here is what changes on other cards and newer drivers.

What stays the same

The pages and rows in this guide exist on GTX 10, GTX 16 and RTX 20, 30, 40 and 50 cards. A few rows appear only when your card or screen supports them (for example Monitor Technology). The recommended values hold for all of them.

Options a newer card or driver may show

The 582.78 screenshots do not have these. Where they exist, leave them at their defaults unless you know why you want them.

Option	Where	Cards	What to do
DLSS (Super Resolution, Frame Generation, Multi Frame Generation, model override)	The game's own menu; overrides in the NVIDIA App (Graphics)	RTX only. Frame Generation: RTX 40 and 50. Multi Frame Generation: RTX 50	Choose it in the game first. Use the NVIDIA App override only for games that lack the newer DLSS version. Frame Generation adds a little input delay: not for competitive games.
Smooth Motion	NVIDIA App, Graphics, Program settings	RTX 40 and 50 (as reported by NVIDIA)	Adds a generated frame in DirectX 11, 12 and Vulkan games. Smoother picture, a little more input delay. Good for slow single-player games.
Image Scaling and its Sharpen slider	Manage 3D settings	All	Off unless you play below native resolution. On an RTX card prefer the game's DLSS.
DSR - Factors with "DL" entries (DLDSR)	Manage 3D settings	RTX	Off for normal play. Turn on only for a sharper picture in a game with FPS to spare.
Resizable BAR	PC BIOS ("Above 4G Decoding" and "Resizable BAR")	RTX 30 and newer, some RTX 20 with a vBIOS update. Not Pascal	On, if your card, motherboard and BIOS support it. The Control Panel's System Information page shows whether it is active.
RTX Video (upscaling and HDR for video)	Video, Adjust video image settings	RTX	For video playback only. It uses more power: leave the default unless you watch a lot of low-quality video.

NVIDIA Control Panel and the NVIDIA App

The **NVIDIA App** is a separate program from NVIDIA Control Panel. It replaces GeForce Experience and holds the per-game settings and the DLSS overrides. NVIDIA announced in 2025 that it was moving some of the most requested 3D settings into the app; the Control Panel is still there and is what this guide uses. Features that need an RTX card are not available on a GTX 10 series card, and a security-only driver such as 582.78 adds no features.

A row you do not know? Match settings by **name**, not by position: rows can move between driver versions. If a row of this guide is missing, skip it. If a new row appears, click it and read the **Description** box at the bottom of the page (visible in the screenshots). When in doubt leave the default.

Windows and monitor checks

The driver is only half of the picture. Check these five places too.

1 Windows refresh rate

Settings, System, Display, **Advanced display**, choose the gaming screen, **Choose a refresh rate**: the highest value. (Tool menu H, "PC health", checks this for you.)

2 Variable refresh rate in windowed games

Settings, System, Display, **Graphics**, **Change default graphics settings**. Turn on **Optimizations for windowed games** (Windows 11). It lets G-SYNC work in borderless and windowed games that use DirectX 10 or 11.

3 Hardware-accelerated GPU scheduling

Same page, **Hardware-accelerated GPU scheduling**. The effect is small and differs per PC. Turn it on, restart, play a game you know. If it stutters more, turn it off again. (This optimizer's tweak 08 sets it.)

4 Game Mode and power mode

Settings, Gaming, **Game Mode**: On. Settings, System, Power, **Power mode**: Best performance (plugged in).

5 The monitor's own menu (OSD)

Turn on **Adaptive-Sync** (some brands call it FreeSync). Leave **Overdrive / response time** on the middle value: the highest setting can add a ghost-like halo. Check that the screen is not in an eco or low-power picture mode.

Two screens

- The screen you game on must be the **main display** (see step 7 of the quick setup).
- A second screen with another refresh rate can make a game stutter. If you see that, close videos and animations on the second screen while you play.
- Use a DisplayPort cable for the gaming screen. Use the cable that came with the monitor if you are not sure.

What gives more FPS than any driver option

- **In-game settings.** Shadows, volumetric effects, view distance and render scale cost far more than any driver option.
- **CPU limit.** A CPU with many cores but a low clock per core limits many games. Lowering the resolution does not help then.
- **Cooling and power.** A hot card or CPU lowers its clocks. Clean dust and check airflow.
- **RAM at its rated speed.** Turn on XMP / EXPO in the BIOS. Tool menu H shows if the RAM runs below its rating.

The driver: update, clean install, roll back

How to change the driver without losing your settings, or your PC.

Update the driver

1. Go to nvidia.com, Drivers, and choose **GeForce**, series **GeForce 10 Series**, product **GeForce GTX 1070**, your Windows version. Download the newest driver the page lists.
2. Run the installer. Choose **NVIDIA Graphics Driver** (and the HD audio driver if you use HDMI or DisplayPort audio).
3. Choose **Custom (Advanced)** only if you want a clean installation (next section). Otherwise **Express** is fine.
4. Restart the PC when the installer asks.

> **Easy Setup and drivers**
> Windows 11 Optimizer never installs a graphics driver for you. Easy Setup only opens the maker's download page when the graphics card has just the basic Windows driver.

Clean installation

A clean installation removes the saved settings of the old driver and starts from the NVIDIA defaults. Use it when you have stutter, crashes or odd colours that started after a driver change.

1. **Take screenshots of your Control Panel pages first.** A clean installation also resets them.
2. In the installer choose **Custom (Advanced)**, press Next, tick **Perform a clean installation**, press Next.
3. Restart, then set the Control Panel again with this guide.

A third-party tool called DDU removes even more leftovers and is run in Safe Mode. It is a last resort for stubborn problems and is not part of this project.

Roll back a driver that caused trouble

1. Right-click Start, **Device Manager**, Display adapters, double-click your GeForce card.
2. Tab **Driver**, button **Roll Back Driver**, follow the steps, restart.
3. If the button is grey: Windows has no older driver stored. Install the previous version from nvidia.com instead, or use the restore point `Before_Driver_Install` from this optimizer's driver tool.

Clear the shader cache

If one game stutters after a driver change, clear the cache once: Start, type **Disk Cleanup**, choose drive C:, tick **DirectX Shader Cache**, OK. The first run of each game afterwards is slower while the cache fills again.

Troubleshooting

Find the symptom in the left column.

Symptom	Likely cause	What to do
"Set up G-SYNC" or "Monitor Technology" is missing	The screen is on HDMI, or it has no Adaptive-Sync, or the option is off in the monitor menu.	Use DisplayPort. Turn on Adaptive-Sync in the monitor menu. Restart the Control Panel.
The picture flickers in menus or loading screens	A screen that NVIDIA has not validated reacts badly to big FPS jumps.	Untick "Enable settings for the selected display model", or cap the FPS in the game.
Tearing is back	FPS above the refresh rate, V-Sync off, or G-SYNC not active for this game.	Style B: V-Sync On in the Control Panel and a cap at refresh minus 3. Check the display mode: fullscreen or borderless.
G-SYNC does not work in a windowed game	"Enable for full screen mode" is selected.	Choose "Enable for windowed and full screen mode", and turn on Windows "Optimizations for windowed games".
FPS is stuck at 60 on a 144 Hz screen	The refresh rate is 60 Hz in Windows or the Control Panel.	Step 4 of "Step by step" and step 1 of "Windows and monitor checks". Set Preferred refresh rate to Highest available.
The game ignores Max Frame Rate	A few games or launchers override it.	Use the game's own FPS limit instead.
Input feels heavy after the changes	V-Sync on without a cap, or Ultra with a game that already has Reflex.	Add the cap (refresh minus 3). Turn Reflex on in the game and use On, not Ultra, in the driver.
The PC idles hot or loud	Power management mode "Prefer maximum performance" keeps clocks up.	Set it back to Optimal power in Global Settings and use Prefer maximum performance in Program Settings of your main game only.
Stutter in one game only	A per-game conflict, or an old shader cache.	Program Settings: set the rows you changed back to "Use global setting". Clear the shader cache (driver chapter).
Stutter in every game after a driver update	Old settings or files that do not fit the new driver.	Do a clean installation (driver chapter).
The screen is black after Apply	The new mode is not supported by the cable or screen.	Wait 15 seconds: Windows goes back by itself. Try a lower refresh rate or another DisplayPort cable.
Control Panel is missing or will not open	It is a separate app on newer Windows installs.	Install "NVIDIA Control Panel" from the Microsoft Store.
Settings were reset after a driver update	The installer ran a clean installation.	Set them again with the cheat sheet on the next page.
Colors look washed out	Output dynamic range is Limited.	Change resolution page: Output dynamic range, Full.

Cheat sheet

Every value of this guide on one page. Print it or keep it next to the Control Panel.

Page / row	Style A (lowest delay)	Style B (smooth)
Adjust image settings	Use the advanced 3D image settings	
Low Latency Mode	On	On
Max Frame Rate	Off	Refresh minus 3 (144 Hz: 141)
Monitor Technology	G-SYNC Compatible	
Power management mode	Prefer maximum performance	
Preferred refresh rate	Highest available	
Shader Cache Size	Unlimited (or 10 GB)	
Texture filtering - Negative LOD bias	Clamp	
Texture filtering - Quality	High performance	Quality
Threaded optimization	Auto	
Vertical sync	Off	On
All other rows of Manage 3D settings	Defaults (Off, Auto or Application-controlled)	
PhysX processor	Your GeForce card	
Change resolution	Native, highest refresh rate, RGB, 8 bpc, Full	
Desktop size and position	Aspect ratio (only if you play below native), Override unticked	
Set up G-SYNC	Enabled, full screen mode, display model ticked, main display	
In the game: V-Sync	Off	Off
In the game: NVIDIA Reflex	On, if offered	
Windows	Refresh rate max, Optimizations for windowed games On, Game Mode On	

Every setting explained

The long version of each Manage 3D settings row. Default = NVIDIA default. Rec. = recommended for gaming.

Setting	What it is and what the options do	Default	Rec.
Image Scaling	NVIDIA's own upscaler (NIS). It renders at a lower resolution and enlarges it. Useful only if you want to play below native resolution. Look: softer.	Off	Off
Ambient Occlusion	Forces extra shadowing in games that support it. Large FPS cost, and only a few older games have a profile for it.	Off	Off
Anisotropic filtering	Keeps textures sharp at a slant (floors, roads). "Application-controlled" lets the game choose; a number forces that level. Forcing 16x costs about 0 to 3% on a GTX 1070 and helps old games that do not offer it.	Application-controlled	Application-controlled
Antialiasing - FXAA	A cheap blur-based edge smoother applied by the driver. Look: blurrier. Many games already have their own.	Off	Off
Antialiasing - Gamma correction	Only affects antialiasing that the driver forces. No measurable FPS effect.	On	On (leave)
Antialiasing - Mode	"Application-controlled" lets the game decide. "Override" forces driver AA, which often does nothing in modern games.	Application-controlled	Application-controlled
Antialiasing - Transparency	Smooths fences and leaves. Supersample modes are expensive.	Off	Off
Background Application Max Frame Rate	Frame cap for a game that is in the background (alt-tabbed). A cap such as 20 saves heat and power; it has no effect while you play.	Off	Off (or 20)
CUDA - GPUs	Which graphics cards CUDA programs may use. Irrelevant for games.	All	All
DSR - Factors	Renders at a higher resolution and shrinks it for a sharper picture. It lowers FPS on purpose.	Off	Off
Low Latency Mode	Limits how many frames the CPU prepares ahead of the GPU. Fewer queued frames = lower input lag. **Off** lets the game decide. **On** limits the queue to 1 frame. **Ultra** sends each frame just in time, with no queue. Works for DirectX 9 and 11 only; DirectX 12, Vulkan and games with NVIDIA Reflex manage this themselves. See the next page.	Off	On (Ultra per game)
Max Frame Rate	A frame cap applied by the driver. See the profiles on the next page.	Off	See profiles
Monitor Technology	G-SYNC Compatible makes the screen follow your FPS (variable refresh rate), so there is no tearing without the delay of V-Sync. Shown only if your screen supports it and it is enabled in Set up G-SYNC.	Fixed Refresh	G-SYNC Compatible (if offered)
MFAA	A cheaper form of multi-sample antialiasing that only works with MSAA on. Rarely useful.	Off	Off

OpenGL GDI compatibility	Compatibility mode for old OpenGL programs.	Auto	Auto
OpenGL rendering GPU	Which card runs OpenGL games. On a PC with one GeForce card, naming it avoids the driver guessing.	Auto-select	Your GeForce card
Power management mode	How hard the card keeps its clocks up. **Optimal power** lowers clocks when it thinks the load is light, which can cause dips in games with changing load. **Adaptive** is similar. **Prefer maximum performance** keeps clocks high while a 3D program runs, which gives more even FPS. Cost: a little more heat and power, and higher idle power on some PCs. If your PC idles hot, set this per game instead of Global.	Optimal power	Prefer maximum performance
Preferred refresh rate	Which refresh rate a game may use. "Highest available" stops a game from picking 60 Hz on a 144 Hz screen. Shown per screen.	Application-controlled	Highest available
Shader Cache Size	How much disk the driver may use to keep compiled shaders. A bigger cache means fewer stutters when a game needs a shader it already used.	Driver default	10 GB or Unlimited
Texture filtering - Anisotropic sample optimization	Takes fewer texture samples. A tiny speed gain, a tiny quality loss.	Off	On (optional)
Texture filtering - Negative LOD bias	Allow = sharper but may shimmer when a game asks for it. Clamp = removes that shimmer. No FPS effect.	Allow	Clamp
Texture filtering - Quality	Quality / Performance / High performance / High quality. Lower settings save about 1 to 3% and lower the picture quality a little. Choose High performance only if you need the last frames.	Quality	Quality (High performance for esports)
Texture filtering - Trilinear optimization	A faster way to blend texture levels. Tiny quality loss.	On	On
Threaded optimization	Lets the driver use several CPU cores. **Auto** lets the driver decide per game. **On** forces it, which can help games limited by one slow CPU core (common on many-core, low-clock CPUs such as Xeons). **Off** fixes rare games that crash or stutter with it. Test per game.	Auto	Auto (try On)
Triple buffering	Only affects OpenGL games with V-Sync.	Off	Off
Vertical sync	Waits for the screen's refresh before showing a frame. It stops tearing but adds input lag and can halve FPS when you fall below the refresh rate. See the profiles on the next page.	Use the 3D application setting	See profiles
Virtual Reality pre-rendered frames	Same idea as Low Latency Mode, for VR headsets only.	1	1
Vulkan/OpenGL present method	"Prefer layered on DXGI swapchain" lets G-SYNC work in windowed Vulkan/OpenGL games. Leave Auto unless you need that.	Auto	Auto

Glossary

The words of this guide in plain language.

Word	Meaning
G-SYNC / Adaptive-Sync / FreeSync	Variable refresh rate: the screen changes its refresh rate to match your FPS, so frames are not torn or delayed. "G-SYNC Compatible" is NVIDIA's name for using a FreeSync or Adaptive-Sync monitor.
V-Sync	Makes the card wait for the screen's refresh before showing a frame. Stops tearing, adds delay when the FPS is below the refresh rate.
Tearing	A horizontal split in the picture where two frames meet.
Frame cap	A limit on FPS, set in the game or the driver.
Input lag / latency	The time between your mouse click and the result on screen.
NVIDIA Reflex	A game option that shortens the render queue. Better than the driver's Low Latency Mode in games that offer it.
Render queue	Frames that the CPU has prepared and the GPU has not drawn yet. A long queue means more delay.
GPU-bound / CPU-bound	Which part limits your FPS. GPU usage at 95% or more means GPU-bound.
Shader cache	Compiled game effects that the driver saves on disk, so they need not be compiled again.
Native resolution	The resolution your screen was built for.
Refresh rate (Hz)	How many times per second the screen updates.
bpc	Bits per color channel. 8 bpc is normal for games and Windows.
DSR	Dynamic Super Resolution: renders above native resolution and shrinks the result.
DDU	Display Driver Uninstaller, a third-party tool that removes driver leftovers.
Pascal	The GPU generation of GTX 10 series cards, such as the GTX 1070.
Clean installation	A driver install that first removes the old driver's saved settings.

Sources and limits

Where the advice comes from, and how far to trust it.

- **Setting names, options and defaults** come from NVIDIA Control Panel itself (the help text under each page, visible in the screenshots) and its well-known behavior.
- **G-SYNC and frame cap:** the advice to enable G-SYNC and V-Sync in the Control Panel, switch V-Sync off in the game and cap a few FPS below the refresh rate is community-tested guidance from Blur Busters, "G-SYNC 101" (blurbusters.com/gsync/gsync101-input-lag-tests-and-settings). It is not an NVIDIA rule.
- **Low Latency Mode:** works in DirectX 9 to 11 games and not in DirectX 12 or Vulkan, according to several hardware sites and forums. NVIDIA Reflex is described on NVIDIA's own Reflex page.
- **Newer cards and the NVIDIA App:** DLSS overrides, Smooth Motion and the move of some 3D settings into the NVIDIA App come from NVIDIA's own news pages (nvidia.com/geforce/news, "NVIDIA App Update Adds DLSS 4 Overrides..." and the global DLSS overrides and RTX 40 Smooth Motion announcement) and Tom's Hardware. Resizable BAR and HDMI details come from general hardware knowledge: check your own card's specification.
- **Driver 582.78:** reported by TweakTown, Guru3D and other news sites as a security-only driver for Maxwell, Pascal and Volta (GTX 750 to GTX 10 series). Searched on 20 October 2026. NVIDIA's own release notes were not available to this guide, so check nvidia.com for the exact notes.
- **DisplayPort and HDMI:** G-SYNC Compatible over DisplayPort 1.2a or newer, and no HDMI variable refresh on Pascal cards, come from hardware sites (Wikipedia, Igor's Lab). Check your monitor's manual.
- **Clean installation:** the steps follow common vendor documentation (for example Dell's support article about NVIDIA clean installation).

> ⚠ **Limits**
>
> Results differ per game and per PC. Treat the recommended values as a good starting point, not a promise. Change one thing at a time and compare the same scene before and after. This guide is not NVIDIA documentation.

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